

brightlearn-Exercise 2 - SQL foundations Ivy-Clarke-02-July-2026

Question 1

01 List all distinct departments in the Student table

```
SELECT DISTINCT department  
FROM Students;
```

department
IT
HR
Finance

Question 2

02 Create ~~and~~ the average age of students per department

```
SELECT department,  
       AVG(age) AS avg-age  
FROM Students  
GROUP BY department;
```

department	avg-age
IT	20.5
HR	22
Finance	23

Question 3

Q3. Show departments with more than 1 student

```
SELECT department,  
COUNT (student_id) AS student_count  
FROM students  
GROUP BY department  
HAVING COUNT (student_id) > 1
```

department	student_count
IT	2
HR	2

Question 4

Q4. Get all students whose age is between 21 and 23

```
SELECT student_id,  
name,  
age,  
department
```

FROM students

WHERE age BETWEEN 21 AND 23;

Student_id	name	age	department
3	Charlie	21	IT
2	Bob	22	HR
5	Eve	22	HR
4	Diana	23	Finance

Question 5

05 List all students in the IT or HR department who are older than 21

```
SELECT student_id,  
       name,  
       age,  
FROM   department  
       students  
WHERE  department IN ('IT', 'HR')  
       AND age > 21;
```

student_id	name	age	department
2	Bob Bob	22	HR
5	Eve Eve	22	HR

Question 6

06 Show total credits per department, only for departments with more than 5 total credits

```
SELECT department,  
       sum(credits) AS total_credits  
FROM   courses  
GROUP BY department  
HAVING sum(credits) > 5;
```

department	total_credits
IT	11

Question 7

07 List all courses that do not have 4 credits

```
SELECT course_id,  
        course_name,  
        department,  
        credits  
FROM courses  
WHERE credits <> 4;
```

course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

Question 8

08 Show the top 3 courses by credits in descending order

```
SELECT course_id,  
        course_name,  
        credits  
FROM courses  
ORDER BY credits DESC  
LIMIT 3;
```

course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

Question 9
Get maximum, minimum, and average grade across all enrollments

```
SELECT  
  MAX(grade) AS max_grade,  
  MIN(grade) AS min_grade,  
  AVG(grade) AS avg_grade
```

FROM enrollments;

max_grade	min_grade	avg_grade
90	78	84.6

Question 10
Count how many enrollments exist per course

```
SELECT  
  course_id,  
  COUNT(enrollment_id) AS enrollment_count  
FROM enrollments  
GROUP BY course_id;
```

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

Question 11

Find total salary and total bonus per department

```
SELECT department,  
       Sum (salary) AS total_salary,  
       Sum (bonus) AS total_bonus  
FROM salaries  
GROUP BY department;
```

department	total_salary	total_bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

Question 12

12. Show departments where average salary is above 55,000

```
SELECT department,  
       AVG (salary) AS avg_salary  
FROM salaries  
GROUP BY department  
HAVING AVG (salary) > 55000;
```

department	avg_salary
IT	61000
Finance	70000

Question 13

13 List employees whose salary plus bonus is greater than 60 000

```
SELECT employee_id,
```

```
       name,
```

```
       salary,
```

```
       bonus,
```

```
       (salary + bonus) AS total_compensation
```

```
FROM salaries
```

```
WHERE (salary + bonus) > 60 000;
```

employee_id	name	salary	bonus	total_compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

Question 14

14 Show total and average budget department. Only include departments with average budget above 70 000

```
SELECT
```

```
       department,
```

```
       sum(budget) AS total_budget
```

```
       avg(budget) AS avg_budget
```

```
FROM projects
```

```
GROUP BY department
```

```
HAVING avg(budget) > 70 000;
```

department	total_budget	avg_budget
IT	270 000	135 000

Question 15

15 List all projects with budgets between 50,000 and 120,000 excluding the marketing department.

```
SELECT  
    project_id,  
    project_name,  
    department,  
    budget
```

```
FROM projects  
WHERE budget BETWEEN 50000 AND 120000  
AND department <> 'Marketing';
```

project_id	project_name	department	budget
1	AI App	IT	120000
2	Payroll system	Finance	80000
3	HR Portal	HR	50000